

SOLUTION REQUIREMENTS SPECIFICATION

Team ID: SWTID-2026-3238 | Project Name: PocketSmart AI | Track: AI-ML & Generative AI

1. Functional Requirements (FR)

- **FR-1 (User Management):** Secure user registration, bcrypt password hashing, and PyJWT-based session authentication.
- **FR-2 (Home Interior Planner):** Budget allocation across furniture, lighting, wall decor, and storage with style filtering (Modern, Bohemian, Minimalist, Luxury).
- **FR-3 (Party & Event Planner):** Headcount-based split across venue, catering, decor, and entertainment.
- **FR-4 (Jewelry & Vision Planner):** Outfit image upload analysis, metal selection, and jewelry pairing advice.
- **FR-5 (GenAI Recommendations):** Gemini 2.5 Flash SDK integration for creative enrichment.
- **FR-6 (Offline Fallback Engine):** Guaranteed local calculations when API keys are unconfigured or unavailable.
- **FR-7 (Provider Search Linking):** Generate query-encoded search links for Amazon, IKEA, Flipkart, Swiggy, Zomato, and OYO.
- **FR-8 (Recommendation History):** Persistent user-specific recommendation logging in SQLite.

2. Non-Functional Requirements (NFR)

- **NFR-1 (Latency):** Local responses < 50ms; live GenAI completions < 2.5s.
- **NFR-2 (Security):** Strict credential hashing, environment variable isolation, and CORS protection.
- **NFR-3 (Reliability):** 100% availability through local fallback handlers.
- **NFR-4 (Usability):** Modern responsive layout functional from 320px mobile to 4K desktop viewports.